Energy accounting and the uncalibrated information bridge

Event-energy accounting

$$2.4 \times 10^{18}\ \text{ATP/event} \times 5.07 \times 10^{-20}\ \text{J/ATP} \approx 0.12\ \text{J/event}$$

Independent accounting estimate; not empirical calorimetry.

κ: event-information bridge

ATP per bit of ignition-event information
Uncalibrated; no adopted numerical value
Matched energy/information measurement required

Landauer: erasure consistency floor

Full reset: ~13–27 bits erased
Canonical partial reset: ~1.0–3.3 bits erased
Erasure bits differ from broadcast-information bits

Paper 1 §4.6; Paper 4 Fig 1. Standard-state ATP conversion; κ and whole-event erasure ratio are distinct.